

Business Rules Implemented by the Code

This document summarizes what the script enforces, regardless of technical details.

1. Data Inputs

- **Banxico** (Banco de México) CSV contains monetary policy rate history.
- **Federal Reserve** CSV contains U.S. federal funds rate history.

Both datasets must contain date identifiers that allow alignment.

2. Banxico Data Rules

1. Skip Metadata

First 12 lines are ignored because they contain headers or notes.

2. Column Renaming

Column SF61745 is always renamed to "Tasa Objetivo".

3. Monthly Data Enforcement

Only rows where the date string starts with "01" are retained.

This ensures exactly one observation per month.

3. FED Data Rules

- No cleanup required; file is assumed to contain valid observations.
- Column `observation_date` is required and must represent a valid date.

4. Merge Rules

1. The FED dataset defines the master timeline for the merged data.
2. Banxico data is attached by joining on row order/index, not by date matching. (This assumes datasets are aligned chronologically.)
3. Only two final columns survive:
 - FEDFUNDS
 - Tasa Objetivo

5. Tick Label Rules

1. Only every second date is shown on the x-axis to maintain readability.
2. All month names are displayed in Spanish abbreviations.
3. Tick labels follow the format:
 - Ene 2023
 - Feb 2024

6. Visualization Rules

1. Both time series must appear on the same plot for comparison.
2. Each marker displays:
 - A numeric value above the point
 - A hover popup showing the rate
3. A dark UI theme is enforced (charcoal background, light text).
4. Plot must export as an HTML file, suitable for execution from a .py script.